

## M4: Homework

● Graded

Student

Jonathan C Elder

Total Points

36.2 / 50 pts

Question 1

4.2.1 (a-e, g, h only)

2.2 / 5 pts

✓ - 0.7 pts e missing

✓ - 0.7 pts g missing

✓ - 0.7 pts h missing

✓ - 0.7 pts d missing

Question 2

4.3.2

5 / 5 pts

✓ - 0 pts Correct

Question 3

4.5.2

4.5 / 5 pts

✓ - 0.5 pts partially correct

Question 4

4.6.1

4.5 / 5 pts

✓ - 0.5 pts G+ partially correct

Question 5

4.8.2

5 / 5 pts

✓ - 0 pts Correct

Question 6

4.9.2

3 / 5 pts

✓ - 0.5 pts b partially correct

✓ - 0.75 pts c incorrect

✓ - 0.75 pts d incorrect

Question 7

4.10.1 (a-e only)

3.5 / 5 pts

✓ - 0.5 pts c partially correct

✓ - 0.5 pts b partially correct

✓ - 0.5 pts e partially correct

Question 8

4.11.1 (a-f only)

5 / 5 pts

✓ - 0 pts Correct

Question 9

4.12.1

3.5 / 5 pts

✓ - 1.5 pts d, e, f not complete

Question 10

4.12.2

0 / 5 pts

✓ - 5 pts answer not provided

Question assigned to the following page: [1](#)

## 4.2 HW

4.2.1 a)  $(1,1) \notin L$  because  $1 \neq 1$  : not reflexive  
 $x < x$  for any  $x$  value : anti-reflexive \*  
 $1 < 2$  but  $2 \not< 1$   $\rightarrow$  : not symmetric  
 if  $x < y$  and  $y < x$  then  $x = y$  : anti-symmetric \*  
 if  $x < y$  and  $y < z$ , then  $x < z$  : transitive \*

b)  $(x, x) \in E$  because  $x \leq x \quad \forall x \in \mathbb{R}$  : reflexive \*  
 $1 \leq 2$  but  $2 \not\leq 1$   $\rightarrow$  : not symmetric  
 if  $x \leq y$  and  $y \leq x$  then  $x = y$  : anti-symmetric \*  
 if  $x \leq y$  and  $y \leq z$  then  $x \leq z$  : transitive \*

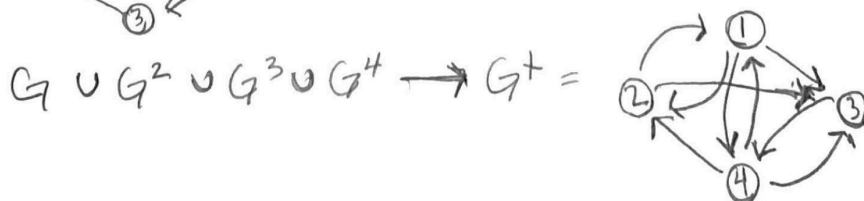
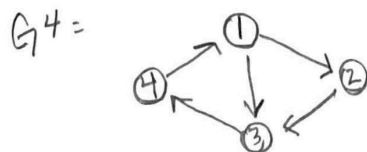
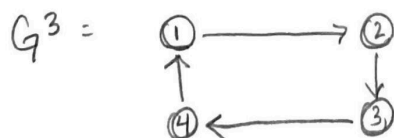
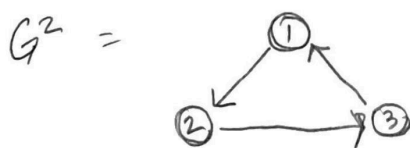
c)  $x^1 = x \quad \forall x$  making it : reflexive \*  
 $2^2 = 4$  but  $4^{1/2} = 2$  and  $\frac{1}{2} \notin \mathbb{Z}^+$  : not symmetric \*  
 If  $x^n = y$  and  $y^n = x$ , then  $(y^n)^n = y$   
 $y^{n^2} = y$   
 so if  $xPy$  and  $yPx$  then  $x = y$   
 because both can not be true. : Anti-symmetric \*  
 If  $xPy$  and  $yPz$ , then  $x^n = y$  and  $y^n = z$   
 $x^{n^2} = y$  and  $y^{n^2} = z$   
 $x^{n^2} = z$  and  $n^2 \in \mathbb{Z} \rightarrow nPz$  : transitive \*

Questions assigned to the following page: [2](#) and [3](#)

### 4.3 HW + 4.5 HW

- 4.3.2
- yes it is a circuit, no it is not a cycle.
  - The longest cycle in the graph is  $d \rightarrow b \rightarrow c \rightarrow g \rightarrow f \rightarrow d$  with a length 5
  - Cycle of length 4 is  $g \rightarrow f \rightarrow e \rightarrow c \rightarrow g$
  - Example of length 5:  $\langle a, b, c, g, f, e \rangle$
  - There are not self-loops in the graph.
  - $\langle a, g, f, c, d \rangle$  is not a walk, trail, or path within the graph because there is no edge from  $f \rightarrow c$ .
  - Yes it is a walk as well as a trail and path.
  - It is neither a circuit or a cycle.

4.5.2 Vertices  $(V) = \{1, 2, 3, 4\}$  Edges  $(E) = \{(1, 2), (1, 4), (4, 2), (3, 4), (2, 1)\}$



Question assigned to the following page: [4](#)

## 4.6 HW

4.6.1 a) The adjacency matrix is

$$G = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$G^2 = G * G$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$G^3 = G^2 * G$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$G^4 = G^3 * G$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$G^+ = G \cup G^2 \cup G^3 \cup G^4$$

$$= G + G^2 + G^3 + G^4$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 2 & 1 & 1 \end{bmatrix} = G^+$$



Question assigned to the following page: [5](#)

## 4.8 HW

4.8.2  $S = \{7, 2, 13, 44, 56, 34, 99, 31, 4, 17\}$

| $X \in S/4$ | Remainder |
|-------------|-----------|
| $7 = 7/4$   | 3         |
| $2 = 2/4$   | 2         |
| $13 = 13/4$ | 1         |
| $44 = 44/4$ | 0         |
| $56 = 56/4$ | 0         |
| $34 = 34/4$ | 2         |
| $99 = 99/4$ | 3         |
| $31 = 31/4$ | 3         |
| $4 = 4/4$   | 0         |
| $17 = 17/4$ | 1         |

$$e_1(0) = \{44, 56, 4\}$$

$$e_1(1) = \{13, 17\}$$

$$e_1(2) = \{2, 34\}$$

$$e_1(3) = \{7, 99, 31\}$$

$$S = e_1(0) \cup e_1(1) \cup e_1(2) \cup e_1(3)$$

$e_1(0), e_1(1), e_1(2), e_1(3)$  are  
they equivalence classes of  $\mathbb{D}$ .

Question assigned to the following page: [6](#)

## 4.9 HW (continued)

4.9.2 a)  $a_{n+1} - a_n = (n+1)^2 - 2(n+1) - n^2 + 2n$   
 $= n^2 + 1 + 2n - 2n - 2 - n^2 + 2n$   
 $= 2n - 1$

$n \geq 1$

Increasing and non-decreasing

b)  $a_{n+1} - a_n = (n+1)^2 - 3(n+1) - n^2 + 3n$   
 $= n^2 + 1 + 2n - 3n - 3 - n^2 + 3n$   
 $= 2n - 2$

$2(1) - 2 = 0$

$2(2) - 2 = 2$

Increasing and non-decreasing

c)  $a_{n+1} - a_n = (n+1)^2 - 4(n+1) - n^2 + 4n$   
 $= n^2 + 1 + 2n - 4n - 4 - n^2 + 4n$   
 $= 2n - 3$

$n=1 \rightarrow 2(1) - 3 = -1$

$n=2 \rightarrow 2(2) - 3 = 1$

Increasing & non-decreasing

d)  $a_{n+1} - a_n = 2^{n+1} - (n+1)! - 2^n + n!$   
 $= 2^n(2-1) - n!(n+1-1)$   
 $= 2^n - n(n!)$

$n=1 \rightarrow 2 - 1 = 1$

$n=2 \rightarrow 4 - 4 = 0$

$n=3 \rightarrow 8 - 18 = -10$

non-increasing and decreasing

e)  $a_{n+1} - a_n = 2^{n+1} - 3^{n+1} - 2^n + 3^n$   
 $= 2^n(2-1) - 3^n(3-1)$   
 $= 2^n - 3^n(2)$   
 $= 2^n - 2 \cdot 3^n$

$n=1 \rightarrow 2 - 6 = -4$

$n=2 \rightarrow 4 - 2 \cdot 9 = -16$

decreasing and non-increasing

Question assigned to the following page: [7](#)

## 4.10 HW

4.10.1 a)  $a_1 = 1$

$$a_2 = 2$$

$$a_3 = a_2 \cdot a_1 = 2 \times 1 = 2$$

$$a_4 = a_3 \cdot a_2 = 2 \times 2 = 4$$

$$a_5 = a_4 \cdot a_3 = 4 \times 2 = 8$$

$$a_6 = a_5 \cdot a_4 = 8 \times 4 = 32$$

The first six terms are 1, 2, 2, 4, 8, 32

b)  $a_1 = 1, a_2 = 5$

$$a_3 = 2a_2 + 3a_1 = 2(5) + 3(1) = 13$$

$$a_4 = 2a_3 + 3a_2 = 2(13) + 3(5) = 36$$

$$a_5 = 2a_4 + 3a_3 = 2(36) + 3(13) = 111$$

$$a_6 = 2a_5 + 3a_4 = 2(111) + 3(36) = 330$$

First 6 terms are 1, 5, 13, 36, 111, 330

c)  $g_1 = 2, g_2 = 1$

$$g_3 = 3 \cdot g_2 + g_1 = 3(1) + 2 = 5$$

$$g_4 = 4 \cdot g_3 + g_2 = 4(5) + 1 = 21$$

$$g_5 = 5 \cdot g_4 + g_3 = 5(21) + 5 = 110$$

$$g_6 = 6 \cdot g_5 + g_4 = 6(110) + 21 = 571$$

first 6 terms of  $g$  are 2, 1, 5, 21, 110, 571

Question assigned to the following page: [7](#)

4.10.1. d)  $c_1 = 4$   
 $c_2 = 5$

$$c_3 = c_2 \cdot c_1 = 5 \cdot 4 = 20$$

$$c_4 = c_3 \cdot c_2 = 20 \cdot 5 = 100$$

$$c_5 = c_4 \cdot c_3 = 100 \cdot 20 = 2000$$

$$c_6 = c_5 \cdot c_4 = 2000 \cdot 100 = 200,000$$

The first 6 terms are 4, 5, 20, 100, 2000, 200000

e)  $b_1 = 1$   
 $b_2 = 3$

$$b_3 = b_2 - 7(b_1) = 3 - 7(1) = -4$$

$$b_4 = b_3 - 7(b_2) = -4 - 7(3) = -25$$

$$b_5 = b_4 - 7(b_3) = -25 - 7(-4) = 4$$

$$b_6 = b_5 - 7(b_4) = 4 - 7(-25) = 179$$

The first 6 terms are 1, 3, -4, -25, 4, 179



Question assigned to the following page: [8](#)

#### 4.11 HW

4.11.1 a)  $\sum_k^4 = -1^k$

$$(-1)^2 + 0^2 + 1^2 + 2^2 + 3^2 + 4^2 \\ 1 + 1 + 4 + 9 + 16 = \boxed{31}$$

b)  $\sum_0^4 = 0^{2^k} = 2^0 + 2^1 + 2^2 + 2^3 + 2^4$   
 $= 1 + 2 + 4 + 7 + 16 = \boxed{31}$

c)  $\sum_k^2 = -3k^3 = (-3)^3 + (-2)^3 + (-1)^3 + (0)^3 + (1)^3 + (2)^3$   
 $= -27 - 8 - 1 + 0 + 1 + 8$   
 $= \boxed{-27}$

d)  $\sum_{k=0}^3 3^k$   
 $= 3^0 + 3^1 + 3^2 + 3^3$   
 $= 1 + 3 + 9 + 27 = \boxed{40}$

e)  $\sum_{k=0}^{200} = 2 + 400 + 60,300$  last term = 602  
 $= \boxed{60,702}$  number terms = 201  
 $\frac{201}{2} [2 + 602]$

f)  $\sum_{k=0}^{200} 2 \times (1.01)^k$   
 $= 2 \sum_{k=0}^{200} (1.01)^k$   
 $= 2(1 + 1.01 + (1.01)^2 + \dots + (1.01)^{200})$   
 $= 2 \left( \frac{1(1.01)^{201} - 1}{1.01 - 1} \right)$   
 $= 2 \left( \frac{7.38917803 - 1}{.01} \right)$   
 $= 200(6.38917803)$   
 $= \boxed{1277.83561}$

Question assigned to the following page: [9](#)

## 4.12 Hwl

4.12.1 a)  $n=3$ , then

$$\sum_{j=1}^3 j^2 = 1^2 + 2^2 + 3^2 = 14$$
$$= \frac{3(3+1)(6+1)}{6} = \frac{3(4)(7)}{6} = 14$$

b)  $P(3)$  is true

$$P(k) = \frac{k(k+1)(2k+1)}{6}$$

c)  $P(k+1) = \frac{(k+1)(k+2)(2k+3)}{6}$

d) for the base case, it must be proven that  $P(1)$  is true.

e)  $k$  is a positive integer. To prove  $P(n)$  will be true if proven that  $P(k+1)$  is true.

f) The inductive hypothesis is that  $P(k)$  is true for a positive integer  $k$ .